

ORIGINAL ARTICLE

Towards Trustworthy Artificial Intelligence for Marine Research, Fisheries and Environmental Management

Jose A. Fernandes-Salvador¹  | Angel Borja¹ | Asier Anabitarte¹ | Igor Granado¹ | Xabier Lekunberri¹ | Yolanda Sagarminaga¹ | Oriol Canals¹ | Anders Lanzen^{1,2} | Mihailo Azhar³ | Jonne Kotta⁴ | Henn Ojaveer⁵  | Anna Spinosa⁶ | Ari-Pekka Jokinen⁷ | Lumi Haraguchi⁸ | Sanjina Upadhyay Stæhr³ | Aritz Pérez⁹ | Iñaki Inza¹⁰ | Sebastian Villasante¹¹ | Gabriela A. Oanta¹² | Catarina N. S. Silva¹³ | Rachel Tiller¹⁴ | Julian Lilkendey^{15,16,17}

¹AZTI Marine Research, Basque Research and Technology Alliance (BRTA), Sukarrieta, Spain | ²IKERBASQUE, Basque Foundation for Science, Bilbao, Spain | ³Department of Ecoscience, Applied Marine Ecology and Modelling, Aarhus University, Roskilde, Denmark | ⁴Estonian Marine Institute, Faculty of Science and Technology, University of Tartu, Tallinn, Estonia | ⁵Pärnu College, University of Tartu, Pärnu, Estonia | ⁶Department of Data Science and Water Quality, Deltares, Delft, MH, the Netherlands | ⁷Finnish Environment Institute (Syke), Helsinki, Finland | ⁸Research Infrastructure, Finnish Environment Institute (Syke), Helsinki, Finland | ⁹Basque Center for Applied Mathematics, Bilbao, Spain | ¹⁰Department of Computer Science and Artificial Intelligence, University of the Basque Country (UPV/EHU), San Sebastian, Spain | ¹¹CRETUS-EqualSea Lab, Oportunus Research Professor, University of Santiago de Compostela, A Coruña, Spain | ¹²Salvador de Madariaga, University Institute of European Studies at the University of A Coruña, A Coruña, Spain | ¹³Centre for Functional Ecology-Science for People & the Planet (CFE), Associate Laboratory TERRA, Department of Life Sciences, University of Coimbra, Coimbra, Portugal | ¹⁴Department of Fisheries and New Biomarine Industry, SINTEF Ocean, Trondheim, Norway | ¹⁵Leibniz Centre for Tropical Marine Research (ZMT), Bremen, Germany | ¹⁶School of Science, Auckland University of Technology (AUT), Auckland, New Zealand | ¹⁷Coastal People: Southern Skies (CPSS), National Centre of Research Excellence, University of Otago, Dunedin, New Zealand

Correspondence: Jose A. Fernandes-Salvador (jfernandes@azti.es)

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ABSTRACT

Artificial Intelligence (AI) is advancing at an unprecedented pace, offering transformative opportunities for marine research, fisheries management, environmental governance and policy development. Particularly in the context of the interconnected data needs of ecosystem management and biodiversity conservation, these technologies can enhance data acquisition, processing and decision support, enabling more integrated approaches to ecosystem management and biodiversity conservation. Yet their adoption in these domains remains limited by the absence of coherent frameworks that ensure transparency, validation and ethical alignment with ecological and socio-economic sustainability goals. This work proposes a comprehensive framework built on three critical pillars for trustworthy AI: socio-economic and legal viability, data governance and technical and scientific robustness. On the one hand it aims to be a guideline for developer teams. On the other hand, it aims to be a guideline for final users (e.g., industry and managers) for designing the requirements and evaluating such systems. The first pillar underscores the need for AI systems that are cost-effective, scalable, environmentally sustainable and legally supported, balancing short-term costs with long-term social and ecological benefits. The second stresses adherence to fair, reliable and ethical access to digital resources, recognising that without strong governance data and algorithms risk becoming fragmented or misused. The third pillar addresses the necessity of rigorous validation across entire AI pipelines, including preprocessing, model evaluation and benchmarking against alternative ground truths, to ensure reliability in real-world applications. Together, these pillars provide a blueprint for developing ethical, reliable and policy-relevant AI systems that can strengthen trust, improve sustainability and guide decision-making across marine science, fisheries, environmental management and European legislation.

1 | Introduction

Artificial intelligence (AI) is advancing at an unprecedented pace, offering transformative opportunities across numerous disciplines and sectors dealing with many of the world's greatest challenges (Stahl and Wright 2018; Dauvergne 2020; Díaz-Rodríguez et al. 2023). In the marine domain, it enables more efficient data acquisition, processing, and decision support in biodiversity research, fisheries management and environmental policy (Rubbens et al. 2023; Kühn et al. 2024; Borja 2025; Reynolds et al. 2024). These developments align with international sustainability objectives, including the United Nations Sustainable Development Goals (UN SDGs) and the Food and Agriculture Organization (FAO)'s Code of Conduct for Responsible Fisheries (Vinuesa et al. 2020; Palomares et al. 2021). Recent advances demonstrate the breadth of AI's potential, from plankton identification and spatiotemporal mapping (Irigoién et al. 2009; Li et al. 2022; Bachimanchi et al. 2024), to habitat and benthic community analyses (Duffy et al. 2019; Marquez et al. 2022; Saleh et al. 2022; Gauci et al. 2016; Mohamed et al. 2018), and species detection and tracking (Silva et al. 2022; Nur Korkmaz et al. 2023; Barroso et al. 2023; Lilkendey et al. 2024). In fisheries research, industrial applications and management, convolutional neural networks can automatically identify species in electronic catch-monitoring images (Tseng and Kuo 2020; Hasegawa et al. 2024), genetic algorithm-based decision tools optimise fishing vessel routes (Granado, Silva, et al. 2025; Granado, Szlapczynska, et al. 2025) and deep learning models classify vessel tracking data to map fishing effort (Taconet et al. 2019; Cheng et al. 2025). Other AI applications include deriving stock abundance indices from sonar (Cabreira et al. 2009; Uranga et al. 2017), automating age estimation from otolith images (Moen et al. 2018; Cayetano et al. 2024), fish recruitment forecasting (Fernandes et al. 2015) and marine spatial planning (Coccoli et al. 2018).

These innovations mark a shift towards autonomous, scalable and non-invasive monitoring systems that deliver more consistent and reproducible results across time and space (Borja et al. 2023). Yet technological progress alone does not guarantee sustainable resource use (Gorelick et al. 2017), equitable access (Borja et al. 2023) or inclusive socio-economic benefits (Banerjee and Duflo 2019). Achieving these outcomes requires robust governance, validation and co-design across science, industry, policy and society. Although AI use in biological sciences has expanded rapidly (Walsh et al. 2021), applications in marine biodiversity monitoring and management remain limited (Borja et al. 2023, 2025). This gap reflects also the absence of frameworks ensuring that AI systems are transparent, validated and ethically aligned with environmental and socio-economic goals (Borja et al. 2023; Díaz-Rodríguez et al. 2023; Fernandes-Salvador et al. 2022). Addressing these challenges motivates this work, which proposes a comprehensive framework for trustworthy AI that integrates marine research, fisheries management, environmental governance and policy development. Uncritical deployment risks reversing intended benefits by amplifying biases, compromising data integrity and eroding trust in scientific outputs (Schwartz et al. 2022). Although European Union (EU) legislation provides a strong foundation for mitigating such risks, these issues are particularly acute in marine and fisheries contexts, where ecosystem complexity and data scarcity make transparency and accountability difficult to achieve (Fernandes-Salvador et al. 2022). High initial costs and limited

technical expertise also hinder adoption, especially among small-scale operators and in developing regions. Persistent challenges include the lack of robust validation frameworks and the difficulty of applying AI effectively in data-limited settings. The EU's Artificial Intelligence Act (AIA) provides a valuable legal foundation (Box 1), yet its interaction with other frameworks will require adaptation to address the specific needs of marine and fisheries applications (Graux et al. 2025). Targeted strategies are needed to strengthen trust, improve access to data and computational infrastructure and ensure that AI development aligns with ecological, socio-economic and regulatory objectives.

AI deployment in marine biodiversity and fisheries offers considerable promise (Borja 2025) but also demands precautionary measures to build confidence among scientists, fishers and policymakers (Sohns et al. 2022). Privacy and ethical safeguards must be in place before deploying technologies aboard vessels (Hartill et al. 2020; Resnik et al. 2025), particularly where cameras or sensors capture data in private or commercial environments. Techniques such as model inversion or membership inference can expose training data or individual identities (Veale et al. 2018), while biases in data or algorithms may impair

BOX 1 | What is AI for European Legislation?

The European Artificial Intelligence Act (AIA, Regulation (EU) 2024/1689) defines an Artificial Intelligence (AI) system as 'a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations or decisions that can influence physical or virtual environments' (*Art. 3(1)*). In the proposal of AIA (COM(2021) 206 final), the following AI approaches classification was used (Figure 1):

- (a) Machine learning (ML) approaches, including supervised, weakly supervised, unsupervised and reinforcement learning.
- (b) Logic- and knowledge-based approaches, including knowledge representation, inductive (logic) programming, knowledge bases, inference and deductive engines, (symbolic) reasoning and expert systems.
- (c) Statistical approaches, Bayesian estimation, search and optimization methods.

While (a) and (b) groups refer to specific fields within AI in computer science, group (c) includes generic fields that are usually considered part of their mathematical or statistical foundation. In the case of statistics, they represent a mathematical science on its own, with the aim of having an open definition of an AI system that is future proof; this classification is neither exhaustive nor exclusive. However, the line where statistics ends and AI starts is unclear since AI foundations are statistical. The final definition of AI is very broad to also accommodate future AI methodologies, but so broad that it could include information technologies (IT) that show enough degree of autonomy, with IT and AI sharing many common opportunities and challenges. In this work we focus on challenges for truthful systems based on machine learning that are useful for other types of AI systems and complex IT systems.

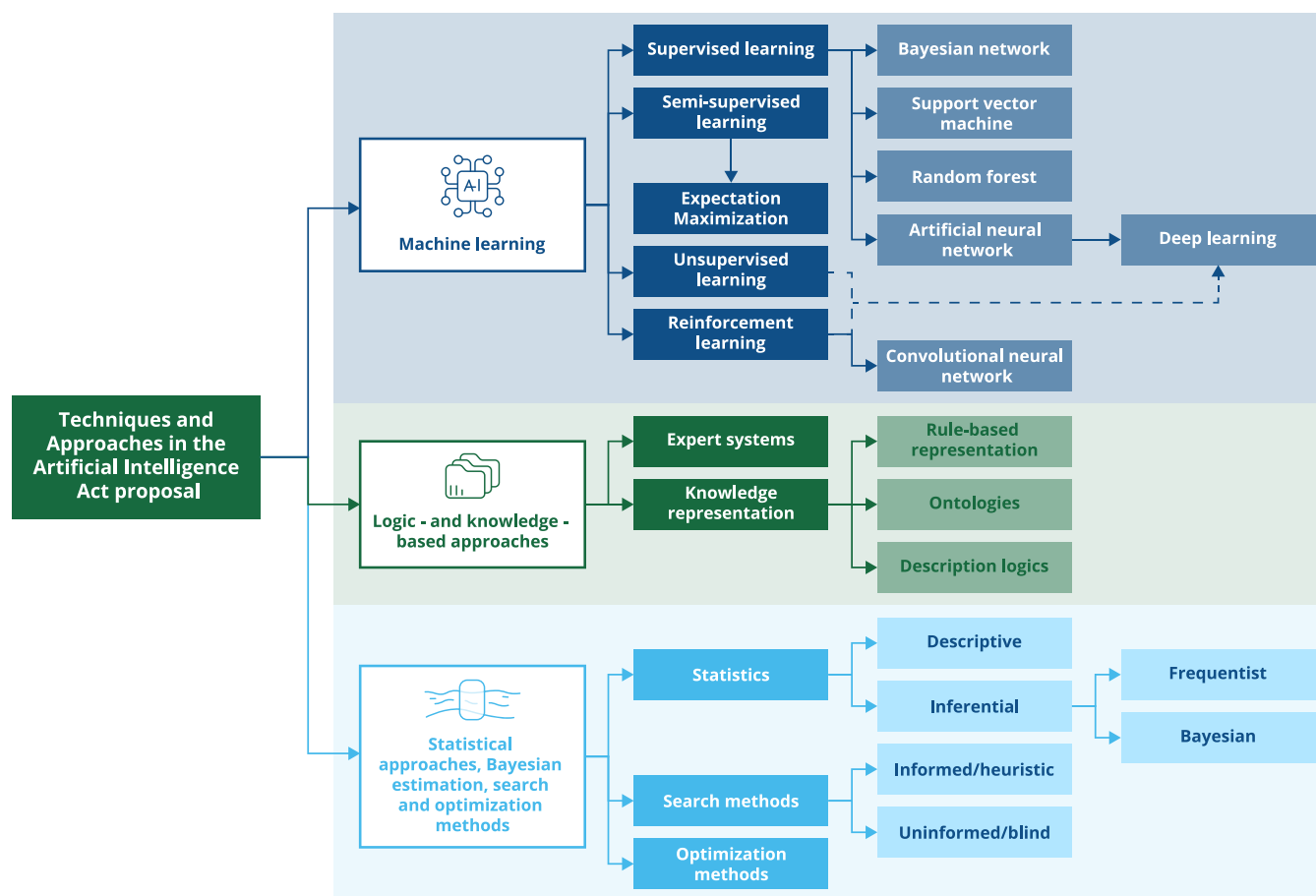


FIGURE 1 | Classification of Artificial Intelligence methods in the AIA proposal, expanded with these commonly used in marine research (adapted from Fernandes-Salvador et al. 2022).

performance in real-world conditions (Probst 2020; Lekunberri et al. 2022; Owen et al. 2025). Moreover, domain shifts—differences between training and deployment contexts—can further distort predictions. Testing AI systems under real conditions, in collaboration with fishers across regions, is essential to strengthen reliability and address perceptions of AI as a ‘black box’ (Sohns et al. 2022). Co-development approaches can thereby enhance transparency and foster acceptance of AI-driven decision-support tools in marine biodiversity and fisheries management.

Here we present a comprehensive framework for fostering ethical, reliable, and policy-relevant AI systems in marine research, fisheries management and environmental governance. By identifying key challenges and structuring them into three pillars of trustworthy AI, this study bridges the gap between the transformative potential of AI and its responsible implementation. Using EU policies as guiding examples, we offer actionable recommendations for strengthening data governance, improving validation practices and aligning AI applications with legal and policy frameworks. The proposed three pillars of socio-economic and legal viability, data governance and technical and scientific robustness illustrate how trustworthy AI can be operationalised within marine research, fisheries and environmental management, and how each pillar is supported by EU legislation.

2 | The Three Pillars of Trustworthy AI

Our study acknowledges that trustworthy AI requires three pillars: (1) socio-economic and legal viability, (2) data governance and (3) robust technical validation according to science state-of-the-art methods (Figure 2). The socio-economic and legal viability pillar acknowledges that the development of AI must consider aspects such as economic viability, resistance to change, legal frameworks, political trust and ethics. The data governance pillar ensures that digital resources (data, models and code) are accessible, reliable and ethically managed. Effective data governance establishes a framework for transparency, accountability and interoperability, addressing critical challenges such as data fragmentation, bias and restricted access. The robust validation pillar requires the use of the state-of-the-art validation schemes (e.g., repeated cross-validation or bootstrapping; Tibshirani and Efron 1993; Rodríguez et al. 2013), the use of statistical tests for comparisons and ground truth testing to provide a robust evaluation. As an example, each pillar is revised in view of European policies (Boxes 2–4).

2.1 | Socio-Economic and Legal Viability Pillar

The first requirement is *economic viability*, which remains a challenge for many fishing companies and associated

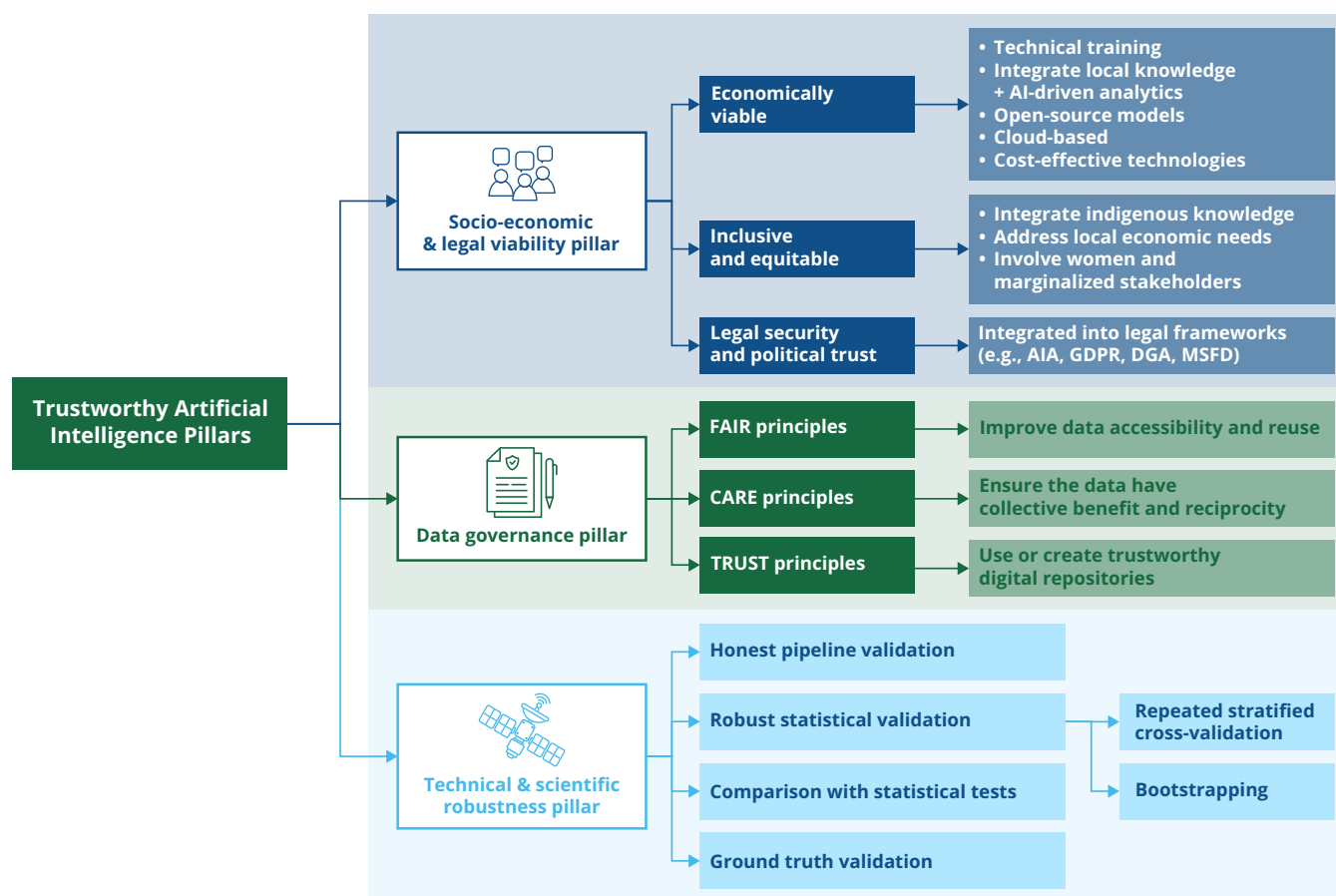


FIGURE 2 | The three pillars for Trustworthy Artificial Intelligence systems with a summary of their main concepts.

industries. The uptake of AI technologies is often constrained by high initial investment costs, limited access to appropriate infrastructure, multidisciplinary experts and a lack of digital literacy within the sector. Many operators perceive such systems as opaque and detached from local ecological knowledge and socio-economic realities (Wing and Woodward 2024). Furthermore, AI systems require continued technical maintenance and updating with an appropriate legal framework. Open-source AI models, cloud-based solutions and cost-effective sensor technologies can provide alternatives to expensive proprietary systems, for example, in fishing vessels digitalization (Uriondo et al. 2024). Additionally, regulators have also historically struggled to balance specificity and flexibility in program designs and contracts, often limiting innovation and adaptation to new technologies (Garren et al. 2021). Addressing these barriers requires targeted efforts to demonstrate the tangible benefits of AI for fisheries-dependent communities where environmental and economic win-wins can be demonstrated (Granado et al. 2024; Goikoetxea et al. 2024; Poisson et al. 2022). Still, there is the fear that higher efficiency does not translate into lower environmental impacts (Chandral et al. 2024) or potential overexploitation (Costello et al. 2008). For example, while AI can improve fuel efficiency and reduce the carbon footprint of fishing vessels (Granado et al. 2021), its wider sustainability implications must be carefully evaluated (Granado, Szlapczynska, et al. 2025; Ghoroghi et al. 2023). However, this is not a new challenge; any technological development must be aligned with sustainable management (Bradley et al. 2019; Sheaves et al. 2020). Finally,

the environmental impact of some of the most computationally expensive AI models depends on the amount of training required and the availability of renewable energy to support their data centres' energy demands (Strubell et al. 2020). Furthermore, computation needs are fostering a new generation of components that are more energy efficient (e.g., Yoo et al. 2025). However, these intensive models are not usually used in small-scale applications like in fisheries and marine research. The most intensive models usually are not trained from scratch but retrained with new data using existing models (e.g., Lekunberri et al. 2022).

AI systems must also be *inclusive and equitable*, to reduce their adoption resistance and potential use for fishing or resources overexploitation. People often exhibit a lack of trust in AI systems because of the aforementioned 'black box' problems associated with them (von Eschenbach 2021; Sohns et al. 2022). Local communities which are dependent on agriculture, forestry, or fisheries, for example, may perceive conservation-oriented AI systems as restrictive and limiting their traditional livelihoods (Gómez-Baggethun and Muradian 2015). Therefore, tools should be designed with participatory co-production of knowledge approaches that integrate local and sector-specific knowledge and expertise and economic needs with scientific and technological development groups that produce these. Historically, technological innovations in fisheries have favoured large-scale industrial operations, exacerbating socio-economic disparities (Bennett et al. 2022). Some of these larger-scale industries

BOX 2 | Ensuring socio-economic and Legal Viability for Marine AI systems.

Socio-economic viability is a cornerstone of trustworthy marine AI, ensuring that benefits outweigh costs while advancing sustainability and social equity. EU legal frameworks provide guidance for integrating economic, environmental, social and legal considerations.

Marine Strategy Framework Directive (MSFD; 2008/56/EC)

Requires sustainable use of marine resources. AI systems should demonstrate contributions to Good Environmental Status (GES) and minimise socio-economic or ecological harm.

General Data Protection Regulation (GDPR; 2016/679)

Safeguards individual rights and equity in data use. Marine AI applications must handle personal and community-linked data ethically and transparently.

European Green Deal (COM(2019) 640 final)

Sets the path towards climate-neutral growth. Developers should account for energy demand, carbon footprint and consistency with decarbonisation goals.

EU Biodiversity Strategy for 2030 (COM(2020) 380 final)

Aims to reverse biodiversity loss and restore ecosystems. AI projects should integrate socio-economic evaluations ensuring that digitalisation supports ecosystem resilience and long-term viability.

Data Governance Act (DGA; 2022/868)

Promotes equitable access to and sharing of data, enabling small-scale fishers and marginalised communities to benefit from marine AI innovations.

Fisheries Control Regulation (1224/2009, as amended by 2023/2842)

Establishes the catch digital information management system for the EU Catch Certification Scheme and mandates fully electronic recording and exchange of catch and landing data to strengthen traceability and combat illegal, unreported and unregulated (IUU) fishing.

Port and flag state control directives (2024a/3099 and 2024b/3100)

Update EU port- and flag-state control regimes, introducing digitalised reporting and data exchange to improve inspection efficiency and transparency.

Artificial Intelligence Act (AIA; 2024/1689)

Requires a risk-management system (Art. 9) and, for some deployers, a Fundamental Rights Impact Assessment (Art. 27). These ensure that high-risk AI applications safeguard health, safety, and fundamental rights, including those of workers.

Nature Restoration Regulation EU 2024/1991

Obliges Member States to use electronic databases and geographic information systems, 'leveraging the opportunities offered by artificial intelligence, advanced data analysis and processing' (Art. 20 § 9).

Example in Practice

An AI system is designed for real-time fish stock monitoring and evaluates its socio-economic and legal impacts by comparing the costs of deployment and maintenance with the economic benefits of sustainable fisheries management, in a setting of co-production of knowledge with representatives of the fishing sector working alongside those from science and from the governance sectors to build mutual trust. Energy efficiency metrics should be analysed to ensure alignment with decarbonization policies, and workforce training programs should be implemented to upskill fishers in using the technology.

Key Recommendations

- Engage the fishing sector in close collaboration in the development phases, with both scientists, technology developers and governance actors to ensure end-user buy-in and mutual trust building.
- Conduct cost-benefit analyses to ensure that the economic and environmental benefits of AI systems outweigh their implementation and maintenance costs.
- Evaluate the environmental impact of AI systems, including energy use and sourcing, to align with sustainability goals (e.g., in decarbonization).
- AI can optimise fishing and associated industries activities to reduce emissions and operational costs.
- Develop workforce transition plans to support shifts from manual to technical roles, ensuring equitable socio-economic outcomes.
- Prioritise equitable access to AI tools and data, ensuring benefits are distributed across all stakeholders, including small-scale and marginalised communities.

have expressed concern that automation might displace jobs, particularly where access to technical training and digital infrastructure is limited (Honarmand Ebrahimi et al. 2021). Small-scale fisheries, which employ over 90% of fishers worldwide and contribute significantly to food security, often lack access to the financial and technical resources necessary to implement AI tools (Bennett et al. 2025). Additionally, gender disparities persist and women are frequently excluded

from decision-making processes and technological training programs (Harper et al. 2020). Thus, AI solutions must be designed within a collaborative context that actively involves marginalised stakeholders in their development and deployment (Wing and Woodward 2024). Transformative technological changes are most effective when integrated into legal frameworks as well, naturally including integrating intellectual property rights and supported by long-term cooperation

BOX 3 | Building Ethical and Transparent Data Governance for Marine AI.

Data governance is the foundation of trustworthy AI, ensuring that data systems are ethical, transparent and interoperable. A combination of European Union policy documents and legal texts provides a strong foundation for ensuring that AI outputs can be developed and validated with verifiable, standardised and traceable data sources.

INSPIRE Directive (2007/2/EC):

Establishes a spatial-data infrastructure enforcing metadata and interoperability standards. These standards ensure that geospatial datasets used to train or test AI models are comparable, traceable and reproducible, directly supporting model training and validation.

MSFD

Mandates harmonised monitoring and reporting of environmental status across Member States. The resulting datasets provide independent ground-truth benchmarks for validating AI-based assessments of marine condition and ecosystem trends.

GDPR

Requires that personal or sensitive data processing be lawful, documented and auditable. This creates data lineage and provenance records that allow validation teams to verify the origin, integrity and lawful re-use of training and testing data.

Open data directive (2019/1024)

Obligates public authorities to make high-value, quality-controlled datasets available for re-use. Such openness enables independent replication and cross-validation of AI findings, a precondition for reproducible science.

European green deal

Promotes data-driven sustainability metrics demanding that environmental AI tools undergo performance validation against measurable efficiency and decarbonisation targets, integrating ecological outcomes into model evaluation.

EU Biodiversity Strategy for 2030

Calls for harmonised biodiversity indicators and transparent monitoring. These consistent indicators supply reference data layers essential for validating AI-derived biodiversity forecasts or habitat-mapping outputs.

DGA

Creates mechanisms for trusted data intermediaries and common data spaces with defined quality standards. These provisions enhance dataset auditability and version control, which are vital for validating AI models over time.

AIA

Requires documentation of data quality, testing procedures, and post-market monitoring for high-risk AI systems. Developers must demonstrate that model performance is reliable, explainable and continuously verified against real-world data.

Port and flag state control directives: Digitalise ship-inspection and compliance records under uniform data standards. These verified datasets enable cross-checking and validation of AI tools used for maritime safety, vessel tracking and risk analytics.

Example in practice

A research team adopts FAIR, CARE and TRUST principles to make their data findable and reusable. By publishing datasets in a GDPR-compliant repository, they ensure privacy while enabling global collaboration.

Key recommendation

- Develop sector-specific guidelines under the AIA to address challenges in marine AI data governance.
- Harmonise these guidelines with GDPR, FAIR, CARE and TRUST principles to foster cross-border collaboration, ethical data management and accountability.
- Use interoperable data platforms and repositories to ensure transparency and accessibility in marine AI research.

and capacity-building efforts, particularly in low-income countries (O'Brien et al. 2024). Moreover, specific gender-responsive policies and training programs can ensure that women in fisheries also benefit from AI applications, particularly in areas such as supply chain optimization and market access (Harper et al. 2020).

Finally, technology must be *supported by legal security and political trust*. The Court of Justice of the European Union in a recent case in the fisheries sector stated that the principle of legal certainty '[...] requires, inter alia, that rules of law be clear, precise and predictable in their effect, especially where they may have negative consequences for individuals and undertakings [...]' (Klohn 2018, C-167/17, EU: C: 2018:833, para. 50; Italy v Council, C-611/17, EU: C: 2019: 332, para. 111). Including relevant regulatory bodies in collaboration with researchers and industry stakeholders is crucial to achieving objectives and ensuring regulations remain effective after

implementation. Fernandes-Salvador et al. (2022) consider that mentions to AI systems could and should be acknowledged in the EU fisheries legislation together with references to digital transformation and new technologies. The Nature Restoration Law, which is the first regulation directly linked to the fisheries sector that has been adopted after the publication of the AIA, is indeed emphasising the adoption of AI when it comes to monitoring. High political trust, as seen in Nordic countries, often reflects effective governance and strong human development (Listhaug and Ringdal 2008). Therefore, involving management, control and surveillance authorities early—alongside researchers and industry—helps foster trust from the outset (Misund et al. 2020). It is considered that the EU's legal framework on the AI is coherent, universal and risk-based and contains strict and well-defined penalties, while States—such as the USA—have adopted legislation in this field following a decentralised and multi-stakeholder model. China moreover uses to some extent the

US model with general guidelines to produce a centralised and comprehensive registration, testing and monitoring framework (Graux et al. 2025) Box 2.

2.2 | Data Governance Pillar

Data governance consists in setting and applying data policies for data gathering, storage, processing and sharing. Appropriate data governance can provide the structure for reliable, ethical and compliant AI by helping to ensure data accessibility (e.g., fair principles), trustworthy data use (e.g., TRUST principles) and people's rights (e.g., CARE principles). From the planning stages of a research project, researchers should consider the types of data and associated metadata to be produced and inform themselves of the vocabularies available in the community, file formats, unit conventions and specific existing repositories (Lin et al. 2020; Kinkade and Shepherd 2022). The use of publicly available media, such as citizen science contributions, raises concerns about consent, attribution and adherence to licensing requirements (Lilkendey 2023). For the AI algorithms, the concepts of explainability and transparency can be achieved with the guidelines on Responsible AI (Fjeld et al. 2020). All these considerations and the principles that will be mentioned later support data governance.

The *FAIR principles*: *Findable* (easy to find with appropriate metadata), *Accessible* (explicit protocols with mechanism for authorised access), *Interoperable* (vocabularies and metadata for automated data merging) and *Reusable* (comply with standards)-were created to improve data accessibility and reuse (Wilkinson et al. 2016; Jacobsen et al. 2020). Data and code uploaded to aggregators or repositories (e.g., Zenodo or Github) ensure data accessibility for the final users. However, these resources may not be useful without continuum evaluation (e.g., Wilkinson et al. 2016). These guidelines were defined to give support to the Open Science movement, adopted by the European Council (Mons et al. 2017). Manso-Narvarte et al. (2024) recommended following the FAIR data principles using the suggestions provided in the OceanObs19 white paper (Tanhua et al. 2019) as well as the use of repositories that allow generating Digital Object Identifiers (DOIs) such as Zenodo or Pangea. Frontiers Media has launched an 'open data' manuscript type in their academic journals, using generative AI to assist the authors in preparing FAIR datasets and metadata and in writing the text accompanying the data, the first manuscript being a 30-year dataset from the Basque coast and estuaries (Borja et al. 2025).

As presented in Table 1, multiple repositories host datasets that comply with the FAIR and/or CARE principles. Furthermore, some repositories have obtained the TRUST certification, which is granted by accredited organisations that validate adherence to the TRUST principles, as illustrated in the second section of the table. In addition, several online tools are available to evaluate the degree of FAIRness of a dataset and to identify specific aspects that require improvement to achieve a higher compliance score. There are also platforms that facilitate the discovery of repositories containing FAIR-compliant datasets, such as [fairs](#)

[haring.org](#) and [re3data.org](#). Both re3data and CoreTrustSeal provide curated lists of repositories that meet the TRUST principles. As observed, there are relatively few repositories that incorporate the CARE principles. This may be attributable to the fact that the subject matter often does not involve highly sensitive data.

The *TRUST principles* aim to go further by creating trusted digital repositories based on the trustworthiness (Lin et al. 2020). TRUST comprises: (1) *Transparency* of the repository and the conditions for use and the ability to manage sensitive data responsibly; (2) *Responsibility* for managing and serving the community by complying with community standards, technically validated, documented, quality control, authenticity protection and long-term persistence; and manage the intellectual property rights of producers and sensitive information protection; (3) *User focus* by making it easier to find, explore and understand the digital resources and monitoring and identifying evolving community expectations and meeting changing needs; (4) *Sustainability* by having a plan well for risk mitigation, business continuity, disaster recovery and succession; secure funding to enable continued use and to maintain desirable data properties; and provide the necessary governance for long-term data preservation so that the data can remain accessible; and (5) *Technology*: A repository should have the technological capabilities to (i) implement relevant standards, tools and technologies for data management and preservation; and (ii) plans and mechanisms in place to prevent, detect and respond to cyber and physical security threats.

The *CARE principles* (*Collective Benefit, Authority to Control, Responsibility and Ethics*) seek to ensure that the data respects the rights of data providers as proposed in an application to indigenous data complementary to FAIR principles (Carroll et al. 2020). This provides a more social vision of the data, highlighting the need for collective benefit and reciprocity, not only for open access.

Table 1 shows examples of systems that allow to check or achieve compliance with these principles. FAIR checkers assist in enhancing the FAIRness of datasets by identifying principles that require improvement, such as interoperability, which often necessitates the incorporation of ontologies and standardised domain-specific vocabularies. Various repositories facilitate the publication of datasets to ensure they are Findable, while TRUST certifications guarantee that these repositories remain reliable over the long term for data access and reuse. Those that follow the CARE principles ensure visible attribution and ownership.

Box 3 provides a summary of the European Union's key policy documents and legal texts which contain the legal requirements for an ethical and transparent data governance in marine science with reference to their needs following the above principles.

2.3 | Technical and Scientific Robustness Pillar

This section summarises four robust validation aspects of AI methods required for trustworthy systems: (1) honest pipeline

validation, (2) robust train-test data splitting, (3) statistical test for comparisons and (4) ground truth validation.

2.3.1 | Honest Pipeline Validation

The development of AI based systems usually consists of several steps starting with data collection, followed by several preprocessing steps and finalising with the model building and results interpretation. Data pre-processing (e.g., discretization algorithms, missing value imputation strategies and feature selection methods) can dramatically improve the final model and its performance (Uusitalo 2007; Fernandes et al. 2010). However, they must be treated as part of the model building and included in the train-test data partition process (Reunanen 2003; Statnikov et al. 2005). This means that data partitioning into training and testing sets must occur before any preprocessing steps are applied, ensuring that all data-driven decisions are learned solely from the training data and evaluated on unseen data (Figure 3). For example, in Fernandes et al. (2010), full processing pipelines are validated to avoid overfitting due to preprocessing optimization.

This concept aligns with closed-loop validation, used in marine energy engineering and fisheries modelling to assess systems within their full feedback cycle. Unlike open-loop testing, closed-loop validation embeds the model in a dynamic feedback loop where its outputs influence subsequent states. In wave energy research, Papini et al. (2024) demonstrated this approach by feeding model-predicted forces back into real-time control, while in fisheries science, Management Strategy Evaluation frameworks apply closed-loop simulations to test management responses under alternative system dynamics (Huynh et al. 2022). Such feedback testing ensures that AI systems are validated across the entire data–decision–outcome cycle.

2.3.2 | Robust Statistical Validation (Train-Test Data Splitting)

The simplest approach for model validation is the hold-out estimator, where a portion of the data is used for training and the remainder is reserved for testing (Larson 1931). However, the performance estimate obtained from a single hold-out split is highly sensitive to the specific data partition (Rodríguez et al. 2009; Roberts et al. 2017). The k -fold cross-validation (Figure 4) provides multiple estimates by splitting the data into k equally sized folds. Each fold is iteratively left out of the training process and used for testing, and the final estimate of performance is obtained by averaging the results from the k -trained models (Lachenbruch and Mickey 1968). Since cross-validation estimates can still exhibit high variance, repeated cross-validation is often used to reduce variance at the cost of higher computational cost. A widely used compromise between robustness and computational efficiency is 10-times repeated 5-fold cross-validation (Bouckaert and Frank 2004). Training-test split used in evaluation should maintain the original dataset's label proportions (stratified), as this practice substantially reduces bias and variance of the

estimate with computational efficiency. An example of how repeated cross-validation can reduce performance overestimation in anchovy recruitment forecasting is provided in Fernandes et al. (2010). Finally, temporal and spatial correlation must also be considered in cross-validation data splitting (Roberts et al. 2017).

2.3.3 | Comparison With Statistical Tests

Model comparisons are often conducted on a single dataset using multiple models or pipelines. Statistical tests provide a framework for assessing whether the observed differences in performance of systems are significant rather than due to random variation using several performance estimates (e.g., from cross-validation, repeated cross-validation or bootstrapping). Statistical tests can be categorised into parametric and non-parametric tests. Parametric tests, such as the paired t -test, assume that the differences in model performance follow a normal distribution. In contrast, non-parametric tests make fewer assumptions about the distribution of the data, making them more robust in practice. While non-parametric tests reduce the risk of misleading conclusions due to incorrect assumptions, they tend to be more conservative, meaning they require stronger evidence to detect significant differences (Santafé et al. 2015; Vehtari and Ojanen 2012). Other approaches are needed if testing over multiple datasets (Singh et al. 2015), for example when testing multiple pipelines to multiple species (Fernandes et al. 2013).

2.3.4 | Ground Truth Validation

Biased training data and overconfidence in expert knowledge are likely the main issues when statistically tested models underperform in real environment (Culverhouse et al. 2003; Sun et al. 2020; Lucas 2020; Owen et al. 2025). Moreover, spurious correlations not reflecting meaningful ecological relationships might be detected (Hill et al. 2020). Therefore, it is advised to conduct additional testing with another type of data or knowledge that can challenge our results beyond train/test data partition. For example, in Taconet et al. (2019), classification of vessels' activity was compared with a database of catches to identify discrepancies. While none of both datasets are all the truth, the inconsistencies allowed to identify some weaknesses of the AI work even if the ground truth has its own biases. Another example is the comparison of on-board fish species classification and size in Lekunberri et al. (2022) with manual estimations at port samplings. Traganos et al. (2022) used the results from localised mapping studies from different parts of the Mediterranean to validate a Mediterranean-wide seagrass mapping process based on Sentinel-2 satellite imagery. In Pucino et al. (2022), AI shoreline estimation models using Sentinel-2 images were validated against a ground truth derived from a combination of surface models from high-resolution UAV imagery and water level estimation using buoy data and tidal modelling. Another example is the introduction of a known number of beds in zooplankton samples (Bachiller et al. 2012). Box 4 details the aid of robust validation in accomplishing policy needs and current policy support.

BOX 4 | Establishing robust validation for marine AI systems.

Robust validation processes are essential for trustworthy AI in marine research, ensuring the reliability, accuracy and transparency of AI systems throughout their lifecycle.

INSPIRE directive

Supports the integration of spatial data, crucial for validating geospatial AI applications in marine research. INSPIRE ensures harmonised and interoperable datasets for model evaluation and benchmarking.

MSFD

Promotes validated environmental models for monitoring marine ecosystems and achieving Good Environmental Status (GES). AI systems used for ecosystem diagnostics must align with these goals through rigorous testing.

GDPR

Emphasises transparency and accountability for data-driven models. Processes must respect data privacy and ensure ethical use of datasets, fostering public trust.

Open data directive

Facilitates access to high-quality, reusable public datasets that are essential for AI validation. Open data sharing complements FAIR principles, enabling broader collaboration and model benchmarking.

European green deal

Highlights the importance of validated AI systems for achieving climate neutrality and addressing environmental challenges. Reliable predictive models support effective decision-making in marine ecosystem management.

EU biodiversity strategy for 2030

Calls for AI-based monitoring tools to track biodiversity and ecosystem health. Robust statistical and ground-truth validation ensures these tools provide actionable and trustworthy insights.

DGA

Strengthens data-sharing practices, ensuring access to trusted datasets for AI validation. The DGA promotes accountability and interoperability in data-driven AI systems.

AIA

Sets out transparency and accountability requirements for AI systems. The AIA mandates robust validation to ensure reliability, particularly for high-risk AI applications affecting people.

Example in practice

A research team developing an AI model for fish stock prediction uses stratified k-fold cross-validation to evaluate model performance across diverse datasets. Ground-truth validation is performed by comparing predictions with field data from port surveys and independent catch records. These steps ensure the model's reliability and alignment with the CFP (Common Fisheries Policy) and MSFD goals.

Key recommendation

Develop sector-specific validation protocols under the AIA, tailored to marine AI applications.

- Harmonise these protocols with GDPR, MSFD and FAIR principles to ensure robust, ethical and compliant validation practices.
- Promote the use of open, interoperable datasets for benchmarking and cross-validation, leveraging EU initiatives like the DGA and Open Data Directive.

3 | Conclusions and Final Considerations

This work highlights the need for robust data governance systems to ensure the ethical and transparent handling of marine data, particularly in fisheries. Validation emerged as another critical area, emphasising the importance of reliable and interpretable AI models that stakeholders can trust. Socio-economic and legal viability highlights the importance of aligning with broader sustainability and equity goals, as well as ensuring co-production frameworks linking to the trust element of validation. Together, these pillars form a holistic trust approach that can guide AI development and use in marine science and fisheries to ensure an effective implementation that can lead to social, ecological and economic changes. Social acceptance and equitable benefits depend on clear value for all actors while respecting their rights which would be supported by the three pillars. Individuals will feel safe using AI systems when there is legal certainty, transparent socio-economic assessments that consider both benefits and costs, ethical, transparent and efficient data handling, and proven technical robustness. These lead also to more efficient systems that often will lead to win-win situations

where costs for industry and environmental impacts can be reduced (e.g., reduction in fuel consumption or more accurate stocks management). The examples discussed throughout this paper—from energy-efficient fishing optimisation over validated stock assessment models to legal security—illustrate how trustworthy AI can deliver tangible benefits when grounded in robust validation and data governance.

A Theory of Change (ToC) is a conceptual framework that articulates how and why a given intervention is expected to lead to its intended outcomes by mapping the hypothesised causal pathways—the sequence of actions and underlying assumptions—that link an intervention to the desired change (Rice et al. 2020). The use of ToC can help identify the conditions under which AI systems deliver equitable and sustainable outcomes for all actors in marine research and fisheries, by making explicit the assumptions and enabling factors (e.g., the three pillars presented here) behind their success (Reinholz and Andrews 2020). Applications of ToC in fisheries research remain sparse but are becoming more common (Contreras-MacBeath et al. 2022; Roscher et al. 2022; Gilman

TABLE 1 | Examples of repositories, certificates and checkers that support FAIR, CARE and TRUST compliance.

Repository	FAIR	CARE	TRUST	Notes
Zenodo (zenodo.org)	X			Open-source repository developed under European OpenAire project.
EmodNET (emodnet.ec.europa.eu)	X			European in situ marine data service focused to taxa available information.
OBIS (obis.org)	X	X	X	Uses WoRMS taxonomic classification and Darwin Core data standard in collaboration with GBIF.
Pangaea (www.pangaea.de ; Felden et al. 2023)	X		X	PANGAEA has a robust interoperability framework that enables sharing metadata and data with registries, portals, and other services
WDC Climate (www.wdc-climate.de)	X		X	Focused on Earth System models data.
DataONE (www.dataone.org)	X			Funded by US National Science Foundation.
ISI-MIP (www.isimip.org)	X	X	X	Environmental projections under climate change scenarios worldwide.
FISH-MIP (fishmip.org)	X	X	X	Fishing projections under climate change scenarios worldwide.
Certificates				
ISO 16363: 2025 (www.iso.org/standard/87472.html)			X	Audit and certification of trustworthy digital repositories by International Organization for Standardisation.
NestorSeal DIN 31644 (www.langzeitarchivierung.de)				An extended self-assessment process for digital archives trustworthy checking.
CoreTrustSeal (www.coretrustseal.org)			X	Certification for repositories comply with over 16 requirements including ISO 16363:2025 and nestor-Seal DIN 31644.
World Data System (worlddatasystem.org)	X	X	X	WDS supports the implementation of repositories based on the certifications.
Checker				
AIR-Checker (Gaignard et al. 2023)	X			Designed to support the findability and reusability of digital life science resources.
Jones and Grootveld (2017)	X			A check list for manual verification.
F-UJI (Devaraju and Huber 2025)	X			An Automated FAIR Data Assessment Tool.
CoreTrustSeal+FAIRenabling Capability Maturity (L'Hours et al. 2022)	X		X	Self-assessment template.

et al. 2024; Rasal et al. 2024), building on earlier examples from conservation research (Garnett et al. 2007; Waylen et al. 2013; Biggs et al. 2017; Balfour et al. 2019). The Three Pillar Framework proposed here provides a pathway for implementing a ToC that links trustworthy AI development to tangible benefits and helps to identify underlying assumptions (Reinholz and Andrews 2020; Rice et al. 2020). Within our ToC, the three pillars function as enabling conditions: socio-economic and legal viability provides the institutional and ethical foundation, data governance ensures transparency and equitable access and technical robustness delivers the scientific credibility required for measurable outcomes. Our pillars activate mechanisms that address risk, enhance transparency and build actors' trust in the long term, as seen

in ToC applications (Rasal et al. 2024), while helping to overcome common barriers (Van Eeden et al. 2021). The resulting outputs are trustworthy, interpretable and usable AI tools that can be confidently integrated into marine science and fisheries decision-making processes.

While the EU has advanced the most comprehensive legal framework for AI to date, non-EU states follow divergent paths that reveal substantial governance asymmetries. In the United States, the Blueprint for an AI Bill of Rights (OSTP 2022) and the Executive Order on Safe, Secure and Trustworthy AI (The White House 2023) rely largely on voluntary guidance and sectoral enforcement rather than binding regulation, leaving environmental and marine applications only loosely covered. India's National

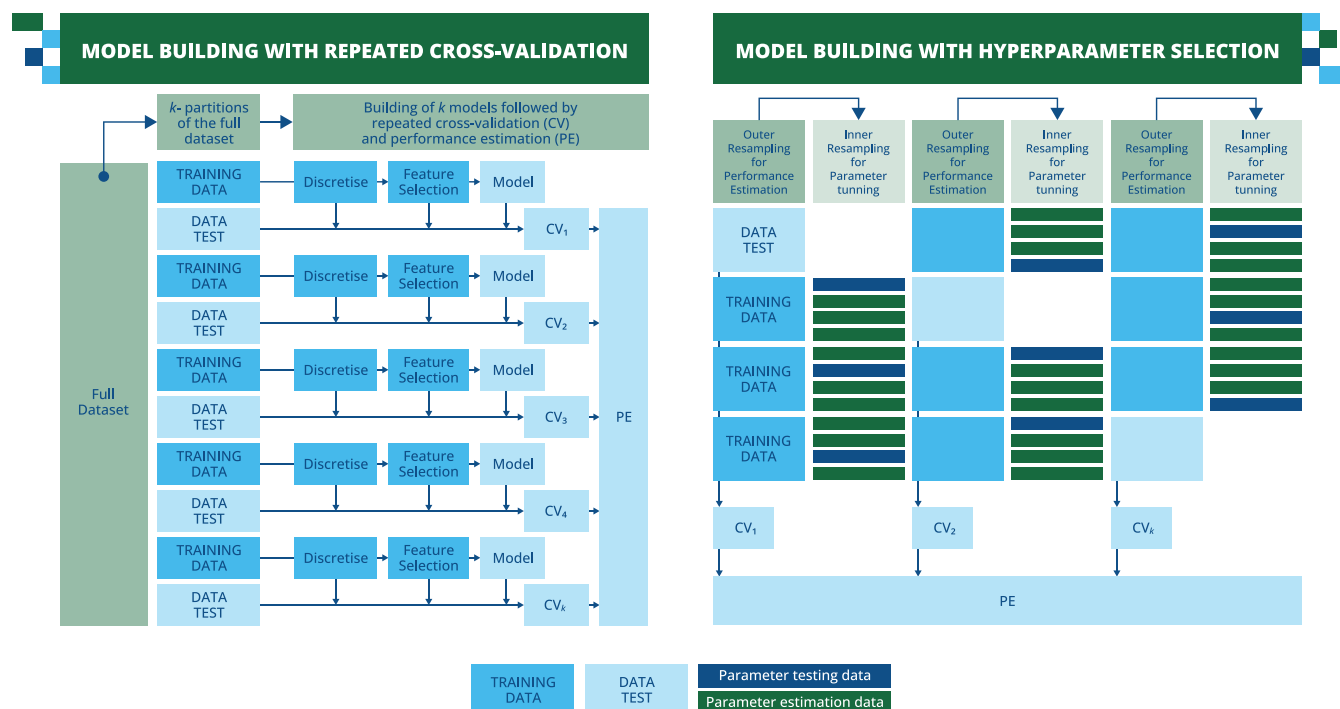


FIGURE 3 | Example of a scheme for validating a pipeline of machine learning methods which include preprocessing steps using repeated cross validation (left). The inner and outer loops are also needed when the model building includes hyperparameter selection for a model (right).

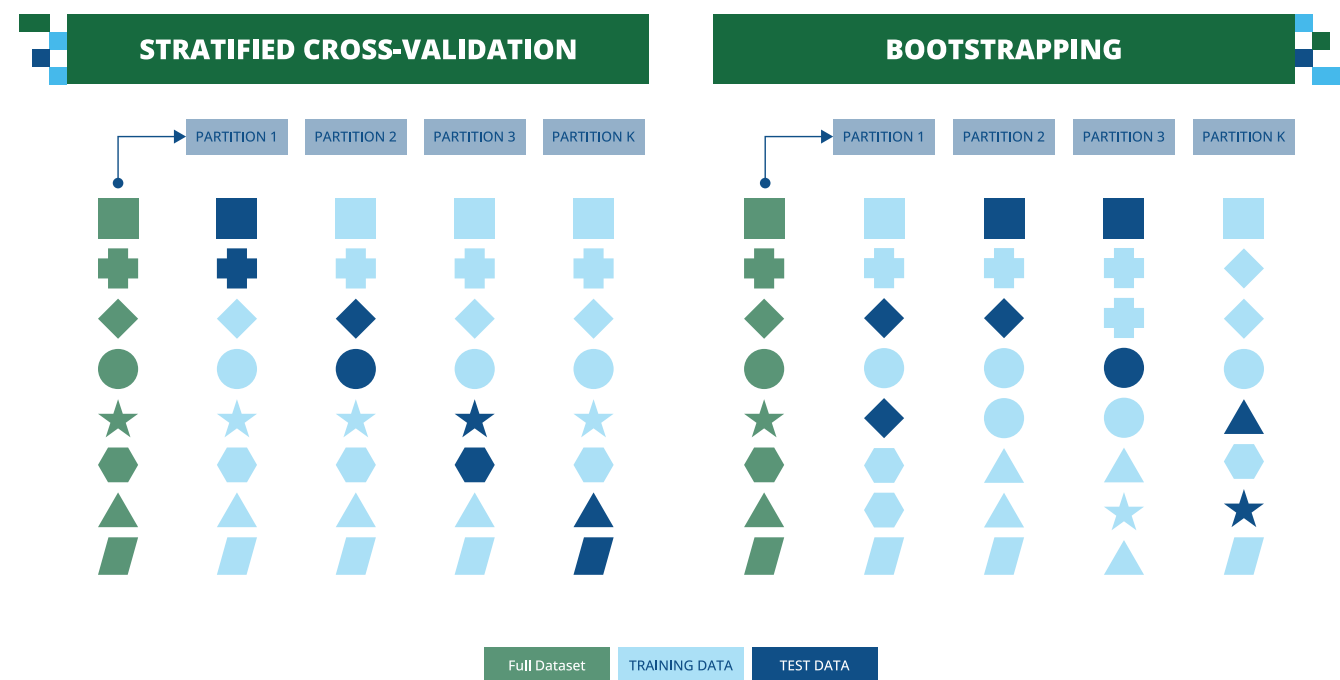


FIGURE 4 | Recommended robust validation schemes. Stratified repeated cross-validation consists of repeating cross-validation (left) several times with random partition of the data. Bootstrapping (right) consists of resampling with replacement and considering the probability of data duplication (0.632) in the performance estimation.

Strategy for Artificial Intelligence (NITI Aayog 2018) promotes ‘AI for All’ but still lacks enforceable safeguards for data governance and environmental accountability, reflecting capacity and infrastructure gaps. New Zealand, by contrast, embeds AI oversight within broader digital-ethics and data-governance frameworks—such as the Algorithm Charter for Aotearoa (New Zealand Government 2020), which emphasise transparency,

accountability and partnership with Māori under Te Tiriti o Waitangi. Although not legally binding, this co-governance model offers valuable guidance for integrating Indigenous data sovereignty and environmental ethics into AI design and validation. China’s New Generation AI Governance Principles (SCPRC 2019) emphasise collective welfare and ecological civilisation, yet operate within a state-centric model where transparency and

independent validation remain limited. These contrasts illustrate that global efforts towards trustworthy AI are fragmented, with differing balances between innovation, oversight and rights protection. For marine and environmental governance, this uneven regulatory landscape complicates cross-border data sharing, validation and ethical accountability—underscoring the value of internationally harmonised principles through bodies such as FAO, IOC-UNESCO and UNEP.

The adoption of trustworthy AI systems in marine research and fisheries aligns with broader international goals, such as the 2030 Agenda for Sustainable Development and UN Ocean Decade which emphasise sustainability, equity and innovation. By adhering to the principles of the Three Pillar Framework, stakeholders can ensure that AI technologies are not only effective but also ethical and inclusive. The emphasis on data governance, validation and socio-economic and legal viability provides a roadmap for harmonising AI advancements with balanced ecological and societal needs. While this framework offers actionable recommendations, its success will depend on cross-disciplinary collaboration and global commitment. As marine ecosystems face mounting pressures from climate change and overexploitation, trustworthy AI systems will play a pivotal role in safeguarding their future. Moving forward, the integration of AI into marine governance should prioritise transparency, inclusivity and alignment with sustainability goals to ensure that the potential of these technologies is fully realised.

Increasing the availability and speed of fisheries and ecological data collection is a win-win for both ecosystem conservation and ecosystem-based fisheries management. While any technological tool can be misused for overexploitation, adopting the three pillars strengthens transparency and reduces this risk. A clear legal framework gives security to developers and users, as well as enforcement capacity to authorities. Strong data governance provides transparency and accountability, which limits opportunities for the misuse of AI systems and gives a competitive advantage to the actors or regions that collaborate. Finally, robust validation practices prevent cherry-picking of results and misleading conclusions about the real capacities of AI systems.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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